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# ----- Enhanced Streamlit Marriage Dataset Dashboard with Background -----

# Import Libraries
import streamlit as st
import pandas as pd
import plotly.express as px
from sklearn.preprocessing import LabelEncoder
from sklearn.cluster import KMeans

# ----- Streamlit Page Config -----
st.set_page_config(page_title="World Marriage Insights", layout="wide")

# ----- Custom Background & Styling -----
st.markdown(
    """
    <style>
    /* Background Image with Gradient Overlay */
    .stApp {
        background: linear-gradient(135deg, rgba(217, 167, 199, 0.7),
        rgba(255, 252, 220, 0.7)),
        url("https://source.unsplash.com/1600x900/?traditional,marriage,culture");
        background-size: cover;
        background-attachment: fixed;
        background-position: center;
    }

    /* Frosted Glass Effect for Components */
    .css-18e3th9 {
        background: rgba(255, 255, 255, 0.8); /* semi-transparent white */
        backdrop-filter: blur(8px);
        border-radius: 20px;
        padding: 20px;
    }

    /* Sidebar Styling */
    .css-1d391kg {
        background: rgba(255, 255, 255, 0.85); /* Adjust opacity */
        backdrop-filter: blur(6px);
        border-radius: 15px;
        padding: 16px;
    }

    /* Text Color Adjustments */
    h1, h2, h3, h4, h5, h6, p, label {
        color: #333333; /* Dark gray */
    }
    </style>
    """
)

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        """
        unsafe_allow_html=True
    )

# ----- Load Dataset -----
@st.cache_data
def load_data():
    # FIXED path with raw string
    df = pd.read_csv("D:\MCA Life 2023-25\College D Y Patil\Sem 4
Project\Major Project Final Submission\World_Marriage_Dashboard_Fixed\World
Marriage Dataset till 2017.csv")

    return df

df = load_data()

# ----- Title & User Guidance -----
st.title("🌐 World Marriage Dataset - Global Trends & Insights (1970-2017)")

st.markdown("""
Welcome to the Interactive Dashboard on Global Marriage Trends.
Here you can explore, analyze, and discover patterns in marriage,
divorce, and other marital statuses across 232 countries over 5 decades.
📌 Tip: Use the sidebar to filter data dynamically by Country, Age
Group, Gender, Marital Status to get personalized insights.
""")

# ----- Sidebar Filters -----
st.sidebar.header("🔍 Explore Data Filters")
countries = st.sidebar.multiselect("Select Countries",
options=df['Country'].unique(),
                                default=df['Country'].unique()[0:10])
age_groups = st.sidebar.multiselect("Select Age Groups",
options=df['AgeGroup'].unique(),
                                default=df['AgeGroup'].unique())
genders = st.sidebar.multiselect("Select Gender", options=df['Sex'].unique(),
default=df['Sex'].unique())
marital_statuses = st.sidebar.multiselect("Select Marital Status",
options=df['MaritalStatus'].unique(),
                                default=df['MaritalStatus'].unique())

# ----- Filter Dataset -----
filtered_df = df[
    (df['Country'].isin(countries)) &
    (df['AgeGroup'].isin(age_groups)) &
    (df['Sex'].isin(genders)) &

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(df['MaritalStatus'].isin(marital_statuses))
]

# ----- Dynamic Insights (Key Stats) -----
st.header("📊 Key Insights from Filtered Data")

if not filtered_df.empty:
    total_records = len(filtered_df)
    married_count = len(filtered_df[filtered_df['MaritalStatus'] ==
'Married'])
    divorced_count = len(filtered_df[filtered_df['MaritalStatus'] ==
'Divorced'])
    most_common_status = filtered_df['MaritalStatus'].mode()[0]
    top_country_married = filtered_df[filtered_df['MaritalStatus'] ==
'Married']['Country'].mode()[0]

    # Stat cards
    col1, col2, col3, col4 = st.columns(4)
    col1.metric("Total Records", total_records)
    col2.metric("Total Married", married_count)
    col3.metric("Total Divorced", divorced_count)
    col4.metric("Most Common Status", most_common_status)

    st.markdown(f"📌 **Top Country for Marriage** (in filtered data):
**{top_country_married}**")
else:
    st.warning("📌 No data available for the selected filters. Please adjust
filters to see insights.")

# ----- Filtered Data Preview -----
with st.expander("📄 Click to View Filtered Data (First 50 Rows)"):
    st.dataframe(filtered_df.head(50))

# ----- Visualization Section -----

# 1. Marital Status Distribution
st.subheader("📊 Marital Status Distribution by Gender")
fig1 = px.histogram(filtered_df, x='MaritalStatus', color='Sex',
barmode='group',
                    title="Distribution of Marital Status", text_auto=True)
st.plotly_chart(fig1, use_container_width=True)

# 2. Top Marriage & Divorce Rates
st.subheader("📊 Country-wise Marriage & Divorce Rates")
country_stats =
filtered_df.groupby('Country')['MaritalStatus'].value_counts().unstack().fill
na(0)

if 'Married' in country_stats.columns:

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top_married = country_stats['Married'].nlargest(10)
st.markdown(f"✔️**{top_married.idxmax()}** has the highest number of
married individuals in the selected data.")
fig2 = px.bar(top_married, x=top_married.index, y=top_married.values,
              title='Top 10 Countries with Highest Marriage Rates',
              labels={'y': 'Number of Married Individuals', 'x':
'Country'}, text_auto=True)
st.plotly_chart(fig2, use_container_width=True)

if 'Divorced' in country_stats.columns:
    top_divorced = country_stats['Divorced'].nlargest(10)
    st.markdown(f"📊 **{top_divorced.idxmax()}** leads in the number of
divorced individuals.")
    fig3 = px.bar(top_divorced, x=top_divorced.index, y=top_divorced.values,
                  title='Top 10 Countries with Highest Divorce Rates',
                  labels={'y': 'Number of Divorced Individuals', 'x':
'Country'}, text_auto=True)
    st.plotly_chart(fig3, use_container_width=True)

# 3. Age Group Analysis
st.subheader("📊 Age Group Analysis by Marital Status & Gender")
fig4 = px.box(filtered_df, x='MaritalStatus', y='AgeGroup', color='Sex',
              title='Age Group vs Marital Status by Gender')
st.plotly_chart(fig4, use_container_width=True)

# 4. Timeline Analysis
st.subheader("📊 Marriage Trends over Time")
fig5 = px.histogram(filtered_df, x='Data Collection (Start Year)',
                    color='MaritalStatus',
                    title='Marital Status over Time', text_auto=True)
st.plotly_chart(fig5, use_container_width=True)

# ----- Clustering Analysis -----
st.subheader("📊 Clustering Analysis: Country, Age, Gender & Marital Status")

# Encode categorical columns for clustering
cluster_df = filtered_df.copy()
encoder = LabelEncoder()

for col in ['Country', 'AgeGroup', 'Sex', 'MaritalStatus']:
    cluster_df[col] = encoder.fit_transform(cluster_df[col].astype(str))

# Check for enough data
if len(cluster_df) > 10:
    kmeans = KMeans(n_clusters=3, random_state=42, n_init=10) # ✔️FIX:
n_init added
    cluster_df['Cluster'] = kmeans.fit_predict(cluster_df[['Country',
'AgeGroup', 'Sex', 'MaritalStatus']])

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fig6 = px.scatter_3d(cluster_df, x='Country', y='AgeGroup',
z='MaritalStatus', color='Cluster',
                      title='3D Cluster Analysis: Country, Age Group,
Marital Status')
st.plotly_chart(fig6, use_container_width=True)

# ----- Footer -----
st.markdown("---")
st.markdown("📌 Note: Dataset from World Marriage Dataset (1970-2017).  
Dashboard created for educational & analytical purposes.")
st.markdown("🏁 Project Analysis Completed - By Nadeem Ahamad")

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